PROJECT BRIEF · RETAIL SITE SELECTION · DATA & AI TOOLING

Transit-Node Site Screening: ranking store locations from public data before money is spent in the field

Berlin case study for a hypothetical 7-Eleven convenience-store pilot · Python, SQLite, Claude API · 241 automated tests, CI on every push · github.com/JuliaZirui/7-eleven-site-selection

1 Client need, current practice and what the tool delivers

Desk research decides where scarce field budget goes, yet today it is slow, unrepeatable and silently over-confident.

Who it serves

- **Retail expansion teams** shortlisting sites in a new city
- **Strategy consultants** in the desk phase of a market-entry engagement
- Both face many candidate sites and budget to visit only a few

Problems with the manual approach

- **Slow:** one competitor inventory for one site took 32 min by hand
- **Inconsistent:** two analysts produce two lists; nobody can rerun the work
- **Missing read as zero:** no data looks like no competition
- **False certainty:** one ranking, no view of how much rests on weak data

What the tool produces, per site

- **Business fit** (0–100) and, separately, **evidence quality** (0–100); never added together
- **Sensitivity analysis** on the least certain inputs, then on weights
- **Differentiation hypotheses**, **evidence gaps** and a **field checklist**
- A sourced report: every figure cites its data snapshot

2 How it works

A fixed, reproducible computation pipeline does the analysis; AI is confined to bounded follow-up and review roles that never change a score.

1 · Ingest Public data as dated, append-only snapshots › **2 · Clean & conflate** Dedupe, merge OSM + Overture, parse opening hours › **3 · AI data audit** Flags implausible records; a person approves fixes › **4 · Metrics** People per day, competitors per day type › **5 · Follow-up agent** Fills missing / stale / conflicting data within a call budget › **6 · Score & stress-test** Fit, evidence quality, sensitivity › **7 · Report** Ranking, gaps, field checklist

CODE MODULE MAP

Layer	Modules (in src/)	Role
Ingestion	`osm_client` `overture_client` `lor_client` `ihk_client` `attraction_client`	Fetch public sources with retries; store raw snapshots
Cleaning	`poi_pipeline` `poi_conflation` `poi_taxonomy` `opening_hours`	Dedupe, merge providers, categorise, parse opening hours
Metrics	`metrics_pipeline` `demand_model` `day_types` `geo`	People per day; competitors open per day type
Calibration	`calibrate_anchors` `calibrate_demand` `calibrate_visitors`	Scale anchors from 139 stations; visitor fit
Scoring & report	`scoring` `report` + Markdown template	Scores, sensitivity; report with citation check
AI components	`agent_followup` `agent_policy` `agent_llm_policy` `audit_*`	Follow-up agent; LLM data audit, human review
Interface & QA	`cli` `site_input` · 241 tests · GitHub Actions CI	Site input, demo; exact rerun checked on every push

DATA LAYER

- **Open sources only:** OSM and Overture Maps (competitors), Berlin MSS 2025 (residents), IHK register (workers), Wikipedia pageviews and VBB timetable (visitors), DB price classes (station tier)
- **SQLite, six tables**; snapshots appended, never overwritten; "data as of" kept apart from "fetched on"; unknown values flagged, never set to 0

ARCHITECTURE

- **Program decides, AI assists:** scoring is deterministic (same inputs, same report, ~1 s, no tokens)
- **Follow-up agent:** triggered by data gaps only; 4 calls per run, 2 per site; a code guard checks every proposal
- **Data audit:** Claude with web search cites sources but cannot edit data; a person accepts each fix

LOGIC IN BRIEF

Element	Method
Demand	People per day within 400 m: residents + workers (verified headcounts) + visitors (≈10 per Wikipedia pageview, fit on 37 attractions); weekday and weekend, 5 : 2
Competition	Convenience-type shops open that day, per 10,000 people
Scoring	0–100 against P10 / P90 of 139 Berlin stations; two pillars, 50 / 50
Evidence quality	Corroboration, freshness, completeness, mapping consistency
Sensitivity	**Inputs first** (visitor conversion, workers, under-mapping, joint downside, absolute counts, day type), **then weights** (scenarios, ±50%, 10,000 draws)
Agent	Refetches or flags data gaps that could change a conclusion; impact-ordered rule by default, Claude optional

3 Case study: three Berlin transit nodes

Brandenburger Tor is the diligence priority. Its lead survives every input stress test, but rests on two figures to verify on site first.

STORE CONCEPT

Fresh food, coffee and convenience goods; 50–100 m² at or next to a station; trading all day, every day. No customer segment or time of day is preset: the question is how strong the competition is at each node and how much business potential the public data show.

WHY THESE THREE SITES

All three share **DB station price class 4**, the same station tier (Zoologischer Garten, class 2, was replaced). What differs is the customer mix, so each tests one profile:

Node	Customer profile	Selection rationale
Brandenburger Tor	Visitor-led city centre	Berlin's best-known landmark station; government district with Bundestag and embassy offices
Warschauer Straße	Office + nightlife mix	Mediaspree offices (e.g. Zalando, Amazon) next to the RAW nightlife cluster
Schönhauser Allee	Residential daily shopping	Prenzlauer Berg shopping street; dense resident catchment

RESULTS (ANALYSIS RUN #26)

Node	People per weekday	Competitors per 10k	Demand / compet. pillar	Business fit	Joint downside*	Evidence quality	Call
Brandenburger Tor	24,806	9.7	88.5 / 100.0	**94.2**	**59.0**	72.9	Prioritise
Warschauer Straße	23,169	27.6	73.2 / 65.1	69.1	43.3	76.4	Lower priority
Schönhauser Allee	14,439	85.9	47.4 / 0.0	23.7	22.7	90.0	Lower priority

* Visitors at the low end of the conversion fit *and* competitor counts scaled up for under-mapping (×2.30 / ×1.60), at the same time. The full report (nine sections, through field checklist and sources) re-checks all 42 cited figures against the database before it is written.

KEY FINDINGS

- **Brandenburger Tor leads both pillars**: high daily demand and the thinnest competition per head of the three.
- **Robust to its weakest inputs:** holds across the visitor-conversion range, worker ranges and absolute counts; with both downsides at once it still leads 59.0 to 43.3. It loses first place only if its competitors are under-counted ~5.9×.
- **Schönhauser Allee is the most contested site:** competitor density above the 90th percentile of the 139 stations; its evidence is the strongest.

WHAT TO VERIFY FIRST

Competition: 27 mapped competitors vs. 66 registered food businesses (41%, vs. 94% at Schönhauser Allee); an on-site walk is item one.
Demand: 57% of weekday demand is estimated visitors and ~6,600 workers are estimated by hand; footfall counts settle both.

4 Manual work vs. the tool

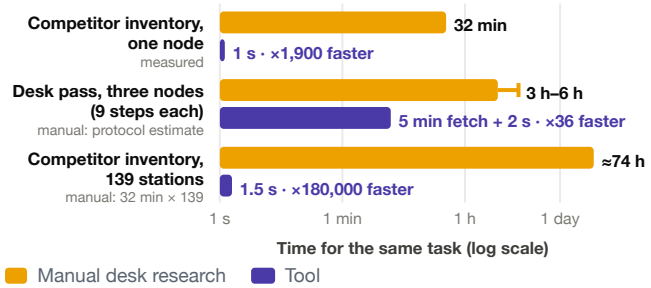
Seconds instead of hours per site, near-zero cost per rerun and city-wide reach: the analyst moves from collecting data to judging it.

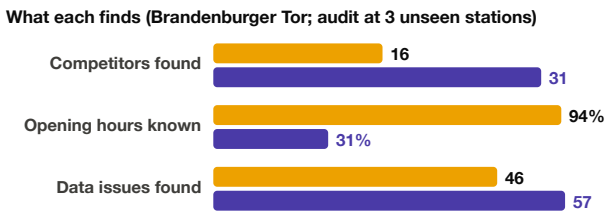
×1,900
32 min → 1 s
competitor inventory for one node

CITY SCALE
74 h → 1.5 s
competitor inventories for all 139 reference stations

LABOUR
9 → 2
analyst touchpoints per node: 9 manual steps become 2 reviews (AI flags, field visits)

COST
€0
marginal cost of a rerun; optional AI audit $0.25–1.25 per site





Issues: manual scan vs. AI flags the analyst confirmed (38 missed by the scan).

One node, one analyst (the tool's author), one non-blind pass; multi-node manual times extrapolated, tool times measured (139 stations from cached data). Manual still wins on opening hours; disagreements became the field checklist.

5 Next steps

- **Field validation at Brandenburger Tor:** competitor walk and footfall counts to test the two figures its lead rests on.
- **Citywide run:** score all 139 reference stations end to end; their core metrics already exist.
- **Configurable store concept:** categories, time windows and weights in a config file, reusable for other small-format retailers.
- **Second manual benchmark** at another node, plus a spot check of near-duplicate merges at the reference stations.
- **Stronger AI checks:** wider audit rules (fresh holdout); LLM extraction of opening hours from shop websites.
- **Hybrid follow-up:** rules decide what to follow up; the LLM only drafts the field question.